

HOW ACCURATE IS GOOD ENOUGH?

Approximating a Civil Engineering Function Using Infinite Series

An Integrated Python Exercise for Civil Engineering Students

Topics Covered:

- Geometric Series
 - Power Series
- Maclaurin Series
 - Taylor Series
- Error Analysis and Engineering Decision-Making

Learning Outcomes:

Students will understand how infinite series can approximate engineering functions, quantify approximation errors, and make engineering decisions based on accuracy requirements.

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PART 1: GEOMETRIC SERIES

Investigation: Compare partial sums with the exact value for $x = 0.5, 0.8$, and 0.9 . How does the number of terms affect accuracy? Why does convergence speed differ?

For $x = 0.5, 0.8, 0.9$, the partial sums converge to $1/(1-x)$ but at very different rates:

x	N=10 error	N=20 error	N=50 error
0.5	9.77e-04	9.54e-07	8.88e-16 (essentially exact)
0.8	4.29e-01	4.61e-02	5.71e-05
0.9	3.14e+00	1.09e+00	4.64e-02 (still not converged)

How does N affect accuracy? Error shrinks every time a term is added, but the rate depends entirely on x .

Why does convergence speed differ? The term added at step k is x^k . When $x = 0.5$, each new term is half the size of the last, so the tail shrinks geometrically fast. When $x = 0.9$, each term is only 10% smaller than the last, so the tail shrinks very slowly — at $x=0.9$ even 50 terms aren't fully converged. In general, convergence speed is governed by how close $|x|$ is to 1: the closer to 1, the slower the series converges.

PART 3: MACLAURIN SERIES FOR $\sin(\theta)$

Investigation: For $\theta = 10^\circ$, calculate approximations using 1, 2, 3, and 4 terms. Compare each to the exact value. What is the error at each step?

Exact $\sin(10^\circ) = 0.17364818$.

N	Approximation	Abs error	% error
1	0.17453293	8.85e-04	0.5095%
2	0.17364683	1.35e-06	0.0008%
3	0.17364818	9.78e-10	0.0000%
4	0.17364818	4.14e-13	0.0000%

Each added term reduces the error by roughly 3 orders of magnitude at this angle — by $N=3$ the approximation is already accurate to 9 decimal places.

PART 4: ENGINEERING INVESTIGATION

1. How does the error change as the angle increases?
2. How does the error change as you add more terms?
3. For small angles ($< 5^\circ$), how many terms are sufficient?
4. For larger angles ($> 20^\circ$), how many terms are needed?

- 1. How does error change as angle increases?** For any fixed N , error grows with angle. At $N=1$, error goes from 0.0051% at 1° up to 4.72% at 30° — nearly a 1000x increase.
- 2. How does error change with more terms?** Error drops sharply — at 30° , going from $N=1$ (4.72%) to $N=2$ (0.0652%) to $N=3$ (0.0004%) to $N=4$ (effectively 0%).
- 3. For small angles ($<5^\circ$), how many terms suffice?** Just 1 term. At 1° and 2° , the $N=1$ error is already 0.005%–0.02%, far under the 0.1% target.
- 4. For larger angles ($>20^\circ$), how many terms are needed?** 2 terms gets you under 0.1% (0.0652% at 30°), but 1 term is not enough (4.72% at 30°).

PART 5: TAYLOR SERIES

Comparison: For the same angles (1° – 30°), calculate errors using:

- Maclaurin (centered at 0°)
- Taylor (centered at 10°)

Which performs better near 10° ? Which is better far from 10° ?

Comparing % error at $N=2$:

Angle	Maclaurin	Taylor ($a=10^\circ$)
1°	0.0000%	8.6093%
2°	0.0000%	3.5633%
5°	0.0000%	0.6331%
10°	0.0008%	0.0000%
15°	0.0040%	0.2974%
20°	0.0126%	1.0261%
30°	0.0652%	3.4822%

Which performs better near 10° ? Taylor — it's exact at $\theta = a = 10^\circ$ by construction (0.0000% error), and stays far more accurate than Maclaurin within a few degrees of 10° .

Which is better far from 10° ? Maclaurin — because it's centered at 0° , it stays highly accurate for small angles (1° – 5°) where Taylor-at- 10° is actually the worse choice (Taylor's error balloons to 8.6% at 1° with only $N=2$, since $\theta - a = -9^\circ$ is a large gap to extrapolate across).

PART 6: ENGINEERING DECISION—ERROR TOLERANCE

Analysis:

1. Does adding more terms always improve the approximation?
2. Why is the Maclaurin approximation most accurate near zero?
3. Why can changing the Taylor expansion point improve accuracy?
4. How does the distance from the expansion point affect accuracy?
5. What is the critical angle for the $\sin(\theta) \approx \theta$ approximation?

Task 1 — minimum terms for <0.1% error:

Angle	Maclaurin N (err)	Taylor N (err)
1°	1 (0.0051%)	4 (0.0207%)
2°	1 (0.0203%)	4 (0.0066%)
5°	2 (0.0000%)	4 (0.0004%)
10°	2 (0.0008%)	1 (0.0000%)
15°	2 (0.0040%)	3 (0.0420%)
20°	2 (0.0126%)	4 (0.0023%)
30°	2 (0.0652%)	4 (0.0299%)

Interesting result: across the full 1°–30° range, Maclaurin never needs more than 2 terms, while Taylor-at-10° needs up to 4 — except right at its own center (10°), where it needs only 1.

Task 2 — critical angle for $\sin(\theta) \approx \theta$: The scan found this 1-term approximation crosses the 0.1% error threshold at $\theta \approx 4.5^\circ$.

Analysis:

1. **Does adding more terms always improve accuracy?** Yes, monotonically, for any fixed angle — $\sin(\theta)$'s Taylor series converges for all real θ . But the rate of improvement slows as N grows (diminishing returns).
2. **Why is Maclaurin most accurate near zero?** Because every derivative used to build it is evaluated at $\theta=0$; the farther θ moves from 0, the more the missing higher-order terms matter to correct for that drift.
3. **Why does moving the Taylor expansion point improve accuracy?** Because the error terms depend on $(\theta - a)$; centering closer to the angle of interest keeps that gap small, so fewer terms are needed nearby.
4. **How does distance from the expansion point affect accuracy?** Error grows with $(\theta - a)$ raised to increasing powers — the farther θ is from a , the faster accuracy degrades for a fixed N .
5. **Critical angle for $\sin(\theta) \approx \theta$:** $\approx 4.5^\circ$, as computed above.

PART 7: FINAL ENGINEERING RECOMMENDATION

Decision Problem:

You are the civil engineer responsible for the calculation. You need an approximation with less than 0.1% error. Would you use the Maclaurin series, a Taylor series centered closer to the angle of interest, or the exact sine function? Defend your choice using your numerical results.

Choice: Maclaurin series, using 2 terms, valid across the full 1°–30° range.

- **Terms required:** 2 (Maclaurin) vs. up to 4 (Taylor-at-10°) to cover the whole range.

- **Percentage error achieved:** worst case 0.0652% (Maclaurin, at 30°) — comfortably under the 0.1% target — vs. Taylor's worst case of 0.0420% at 4 terms.
- **Convergence behavior:** both converge for all real θ ; the practical question isn't whether they converge but how few terms are needed for the angles actually used.
- **Computational simplicity vs. accuracy tradeoff:** a fixed 2-term Maclaurin polynomial is cheap (no trig call) and needs no angle-dependent re-centering, unlike Taylor which only wins near its own expansion point.
- **Valid angle range:** 1° – 30° , with 2 Maclaurin terms meeting $<0.1\%$ error throughout.

Taylor-at- 10° would only be the better pick if the real-world use case concentrated angles tightly around 10° — here it's not, so Maclaurin wins on both simplicity and coverage. The pure small-angle approximation $\sin(\theta) \approx \theta$ (Task 2) is explicitly ruled out beyond $\approx 4.5^\circ$, so it can't stand in on its own for this range.